Skills Summary

Solving Radical Equations and Checking Extraneous Roots

Use this reference while completing your worksheet or when studying.

Skill 1 — Isolate, then square (isolate-and-square)

Method: (1) get the radical alone on one side; (2) square both sides; (3) solve what is left; (4) substitute every candidate into the *original* equation.

Squaring an un-isolated radical keeps a radical ($(\sqrt{x}+2)^2 = x + 4\sqrt{x} + 4$), which is why step 1 comes first.

Example: Solve $\sqrt{x+2} = 4$.
Step 1. The radical is already alone and the right side is positive, so squaring is safe and undoes the square root.
Step 2. $x + 2 = 16$, so $\boldsymbol{x = 14}$. Check: $\sqrt{14+2} = \sqrt{16} = 4$.

Try it: Solve $\sqrt{2x-1} = 5$.

check: $\boldsymbol{x = 13}$

Skill 2 — Reject extraneous roots (check-extraneous)

Why they appear: squaring is not reversible. From $a = b$ you may conclude $a^2 = b^2$, but $a^2 = b^2$ only gives $a = \pm b$, so the squared equation can gain roots.

Fast filter: $\sqrt{\ } \geq 0$ always. If a candidate makes the non-radical side negative, it is extraneous — no arithmetic needed.

Example: Solve $\sqrt{x+3} = x - 3$.
Step 1. The right side can be negative here, so squaring may invent a root and every candidate must be tested in the original equation.
Step 2. $x + 3 = x^2 - 6x + 9$ gives $x^2 - 7x + 6 = 0$, so $x = 1$ or $x = 6$. At $x = 1$ the right side is $-2 < 0$, so it is extraneous; $x = 6$ checks: $\boldsymbol{x = 6}$.

Try it: Solve $\sqrt{x+1} = x - 1$ and say which candidate is extraneous.

check: $\boldsymbol{x = 3}$

Skill 3 — Radical and rational-exponent form

Translation: $x^{m/n} = (\sqrt[n]{x})^m = \sqrt[n]{x^m}$ — the denominator is the root, the numerator is the power.

To solve $x^{m/n} = k$, raise both sides to the reciprocal power n/m. Take the root *first* when evaluating: the numbers stay small.

Example: Evaluate $27^{2/3}$.
Step 1. The denominator three tells me to take a cube root, and the numerator two tells me to square, so I do the root first.
Step 2. $27^{2/3} = \left(\sqrt[3]{27}\right)^2 = 3^2 = \mathbf{9}$

Try it: Solve $x^{3/2} = 27$ (raise both sides to the power $2/3$).

check: $\boldsymbol{x} = \mathbf{9}$

Skill 4 — Applied radical models

Method: substitute the given value, solve the radical equation, then ask two questions: does the answer check in the model, and does it make sense for the quantity (a length, a time, and a speed are never negative)?

Keep units with the answer; the model's variables are quantities, not just letters.

Example: Free fall $t = \sqrt{h/16}$ (t in seconds, h in feet). Find h when $t = 2$.
Step 1. The unknown sits under the radical, so I substitute the given time and solve the radical equation for the height.
Step 2. $2 = \sqrt{h/16}$, square: $4 = h/16$, so $\boldsymbol{h} = \mathbf{64}$ ft. Check: $\sqrt{64/16} = \sqrt{4} = 2$.

Try it: Skid model $s = \sqrt{24x}$ (s in miles per hour, x in feet). Find x when $s = 30$.

check: $\boldsymbol{x} = \mathbf{37.5}$ ft

(!) Watch out: An answer is not finished at the solving step. Check every candidate in the ORIGINAL equation, not in the squared one — the squared equation accepts extraneous roots by construction.